

human cord

An Infrastructure Where Lies Cannot Hide

A Cryptographic Infrastructure for Documentary Integrity in the AI Era

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1. Abstract

As the capabilities of generative AI advance, the cost of superficially reproducing official documents and certificates is falling rapidly. Conventional digital signatures and public-key infrastructure (PKI) can detect tampering, but no adequate mechanism exists to give a document the persistent, intrinsic property that "**this document could only have been created by its issuer.**" The human cord proposed in this white paper is a concept for an integrated cryptographic infrastructure that fills this gap.

human cord integrates **10 pillars** into **five domains plus one cross-cutting layer**: (i) the macro layer (the egg chain), (ii) the micro layer (the contents of each egg), (iii) the key-derivation layer, (iv) the verification layer, (v) the active-response layer, and (vi) the cross-cutting failure-boundary layer. Each pillar is, on its own, strongly connected to existing research and standards; rather than reinventing the wheel, the novelty lies in how they are combined. We present three principal contributions: first, a Subliminal Channel extended to the document level (Pillar 6); second, self-observation resistance that rigorously separates "form" from "secret" at the failure boundary (Pillar 10); and third, the tally operators `+/-` (Pillar 3), which reformulate BLS aggregate signatures as operators and branch automatically on verification failure.

As mid-to-long-term standardization goals, we have ISO/IEC 18033 (encryption algorithms), the annexes of ISO/IEC 27002 (controls), and IETF RFCs (via the IRTF CFRG) in view. At the time of writing, Phase 0 (design documentation) is complete, and in Phase 1 (a minimal Node.js POC) all ten pillars have reached implementation.

2. Motivation: Why a Mechanism, Not People, Should Guard the Truth

2.1 The Problem We Set Out to Solve

The outward appearance of official documents and certificates is becoming easy to reproduce by combining large language models with image-generation models. A digital signature is sufficient to show that "the issuer signed at a given point in time," but it remains passive — it can detect later modification, yet does not endow the whole document with the active property that **"this document could only have been created by its issuer in the first place."** Blockchains are strong at making public records tamper-resistant, but they are not a technology for embedding issuer-identity inside the document itself; the two should be complementary.

2.2 The Project Constitution (Three Principles)

human cord holds, prior to any technical design, three principles as its constitution. First, build a **mechanism where lies cannot hide**: rather than preventing tampering outright, prioritize that the moment tampering occurs, a trace remains and persists. Second, create **a state in which rules are observed, rather than asserting rights**: instead of "rights-assertion" mechanisms such as access restriction, copy protection, and automatic reporting, aim for "rule-compliance" infrastructure in which anyone can judge authenticity. Third, **AI is AI and people are people** — rather than fighting, each fulfills its own role.

2.3 The Fourth Article: Separation of Technology, Law, and Ethics

Technology devotes itself to preserving truth; the role of punishing malice is left to law, and the role of choosing what is right is left to ethics. human cord merely raises smoke; judging that smoke belongs to the domain of people and law. We make explicit, as a premise of our design decisions, that technology **must not replace** law or ethics.

2.4 Three Gaps Existing Countermeasures Leave Open

First, even when tampering can be detected, it is hard to say strongly *from the document itself* "whose it is" (the limit of PKI). Second, cryptographic evidence is lost across media round-trips such as printing, scanning, and photography. Third, internal structure is exposed at the moment of failure (the lack of a systematic countermeasure against side-channel and fault attacks). The ten pillars presented here address these three gaps head-on through Pillars 4, 7, and 10 respectively.

3. Architecture Overview

The architecture of human cord can be grasped as a whole through the metaphor of an "egg chain" flowing over a transmission path. Each egg is sealed with standard authenticated encryption (AEAD),

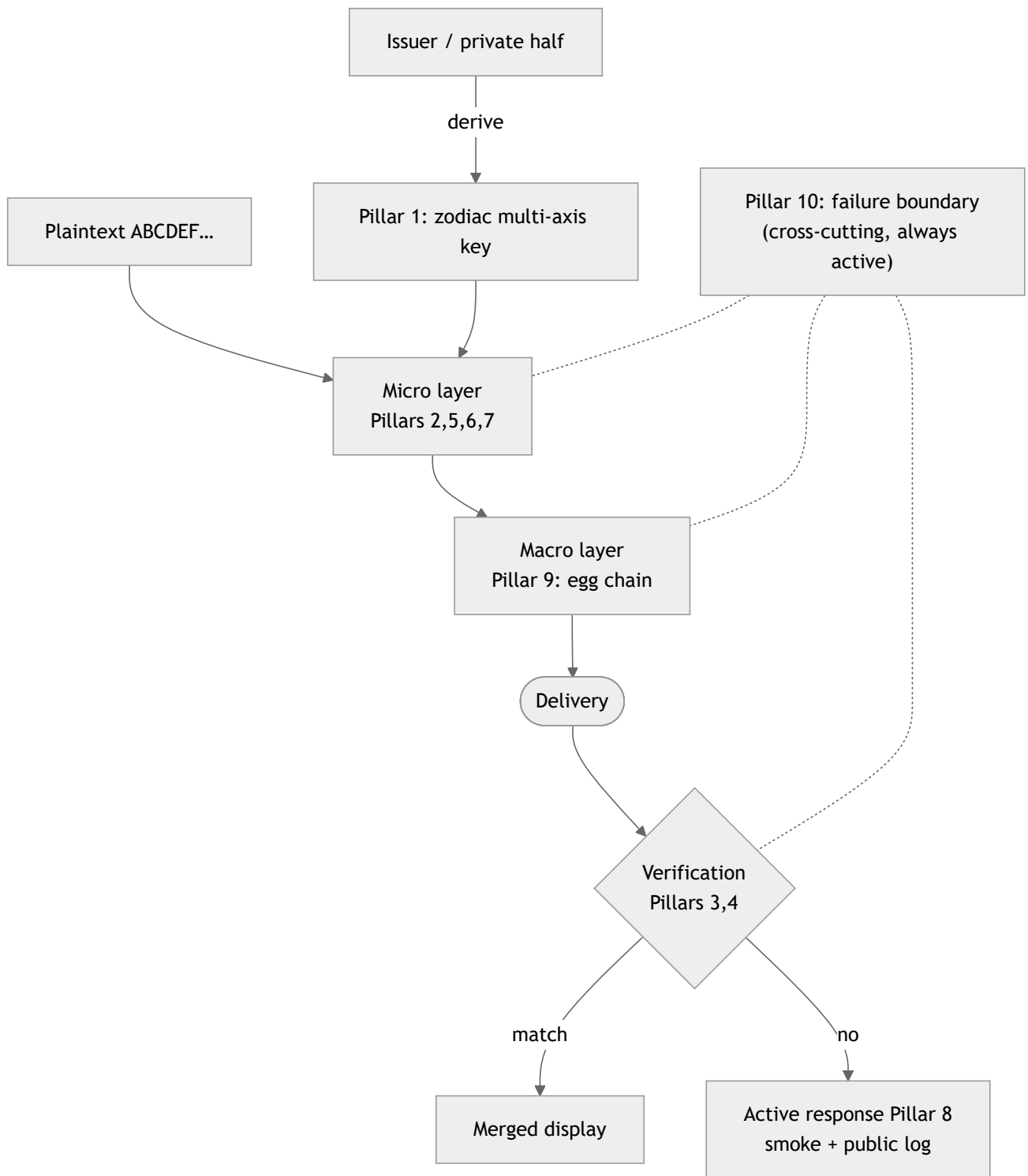
and the eggs are linked to one another by a hash chain (the macro layer). The contents of each egg are composed of multiple layers — eye-glyphs, landscape, latent noise, and a physical-layer signal (the micro layer). Over these, three further layers — key derivation, verification, and active response — are placed from the outside, and finally a cross-cutting behavioral discipline we call "self-observation resistance at the failure boundary" runs vertically through the entire process. This chapter surveys the role of each domain.

3.1 Five Domains Plus One Cross-Cutting Layer

Domain	Pillars	Role
Macro layer (egg chain)	9	A carrier riding on standard AEAD
Micro layer (egg contents)	2, 5, 6, 7	Eye-glyph substitution / landscape blending / moving core / covert channel / physical-layer signal
Key-derivation layer	1	Zodiac-style multi-axis key = time (public) × issuer secret (private)
Verification layer	3, 4	Tally operators <code>+/-</code> + the issuer's private half
Active-response layer	8	Tamper detection → smoke + public log
Failure boundary (cross-cutting)	10	Runs vertically through the whole process. Desync detection / graceful degradation / internal-state leak resistance / observation evasion

3.2 Architecture Diagram

Figure 1 shows a simplified data-flow diagram (a rendered version is at `docs/figures/architecture.en.svg`).



3.3 The Ten Pillars at a Glance

#	Pillar	One-line summary
1	Zodiac-style multi-axis key generation	A moment-specific codebook from time + secret
2	Landscape blending	Only eye-glyphs are substituted; the rest passes through
3	Tally operators	+ to merge / - to fire a difference
4	The issuer's private half	Without the issuer secret, neither verification nor decryption is possible
5	A living operator	Internal state keeps advancing via a ratchet
6	Subliminal channel	Noise in the eye-glyphs, decryptable only by the issuer
7	Physical-layer cross-modal signal	Detectable by phone sensors/cameras; survives printing
8	Active tamper response	On detection, raise "smoke"
9	Egg-flow architecture	AEAD + hash chain
10	Self-observation resistance at the failure boundary	Cross-cutting. On break, only form leaks; the secret does not

3.4 Two Layers of Verification: Issuer-Mediated + Public

human cord carries two verification modes together.

- **(i) Issuer-mediated verification (symmetric core):** Pillar 9 eggs (AEAD) + Pillar 4 issuer secret. Only the issuer can decrypt and verify; tampering is detected while the body stays hidden. The issuer-only covert channel (Pillar 6) belongs to this layer.
- **(ii) Public verification (public-key layer, Ed25519 / RFC 8032):** the issuer signs the "publicly disclosable facts," and **anyone can verify offline with only the issuer's public key** (no issuer secret needed). This fits cases like certificates where "the facts are public and a third party confirms authenticity."

Both modes compose: a single issuance can simultaneously achieve "only the issuer hides/reads / anyone verifies authenticity." Public verification runs identically under the browser's Web Crypto, so third-party verification completes on a **static verification page without a server**. This two-layer structure lets confidentiality (issuer-mediated) and public authenticity-checking (public verification) — requirements that often conflict — be selected and combined per use case.

3.5 Layering Toward Standardization: Core and Extensions

The ten pillars are a conceptual map of the whole design, but they are not uniform from a standardization or implementation standpoint. Taking each pillar through a "threat → mechanism → property" teardown sorts them, by where the guarantee lives and how mature it is, into the following four layers. This layering exists to frame standardization realistically as a **minimal core → public verification → extensions** staged process, rather than a single adoption of all ten pillars (§ 6.2).

Layer	Pillars included	Positioning
(I) Symmetric issuing core (minimal kernel)	1 key derivation + 9 AEAD chain + 5 key advance + 10 minimal disclosure	An inseparable single seal/open core (cord. js / egg. js). Standardization targets this minimal kernel first
(II) Public-verification layer	The public-key layer of Pillar 4 (Ed25519 / RFC 8032)	The natural fit for long-lived, public, repeatedly verified documents (certificates, etc.). § 3.4(ii)
(III) Distinctive value (extensions of the core)	3 confidential matching / 4 threshold issuance (Shamir) / 7 carrier codec (Reed-Solomon physical transport)	Differentiating features that public-key signing alone cannot provide
(IV) Future extensions, cross-cutting principle, operational conventions	4 Fuzzy Extractor · 6 Subliminal · 7 replay prevention (use-case-dependent future extensions) / 10 (a design principle imposed across the whole pipeline = an invariant, not a standalone feature) / 8 smoke (recording into a tamper-evident audit log = an operational convention an adopter's existing audit infrastructure can absorb) / 2 landscape blending (a verification-independent display layer)	Standardized and implemented incrementally per use case and maturity

The key point is that layers (I)–(III) constitute the **substantive core of the specification (roughly five functions)**, while layer (IV) separates into "future extensions," "a principle imposed on the whole," "operational conventions," and "a display layer." This does not remove any of the ten pillars; it **clarifies the order and granularity in which they are written as a specification**. The ten pillars as a conceptual map (the metaphor ↔ cryptographic-engineering correspondence in Appendix A) are preserved as is.

4. Principal Contributions

4.1 Pillar 6: A Subliminal Channel Extended to the Document Level

The concept of the subliminal channel was proposed by Gustavus Simmons at CRYPTO '83; it makes it possible to embed, inside a digital signature, a second message that only the signer can read out [1]. To a verifier the signature looks entirely ordinary, and only a holder of a particular key can decrypt the hidden message. The academic literature is rich — the existence of a subliminal channel in Bitcoin's ECDSA has recently been pointed out — yet **no practically deployed standard exists at present**.

human cord scales this concept up from a single digital signature to **the level of an entire document**. The minute noise attached to eye-glyphs (selectively substituted characters introduced in Pillar 2) accumulates across the whole document and constitutes a "hidden document" decryptable

only by the issuer. To a verifier's eye it displays as an ordinary certificate, while the issuer, by reading out the covert channel, can internally render the judgment "this is indeed something I issued."

The significance of this pillar is that even if AI achieves a perfect reproduction of the outward appearance, the covert channel itself cannot be generated without the issuer secret. In other words, against a situation in which visual authentication is losing trustworthiness in the AI era, it makes it possible to **retain issuer-identity in an invisible layer**.

Operational positioning (an important reservation): a subliminal (covert) channel also has a property that audit and standardization contexts often treat with suspicion — as something to be detected and eliminated. This pillar is therefore **excluded from the verifiable core (§ 3.5 layers (I)–(III)) and quarantined as a use-case-limited future extension (layer (IV))**. Neither issuer-mediated verification (§ 3.4(i)) nor public verification (§ 3.4(ii)) depends on it in any way. Adopters should judge case by case whether the mere existence of a covert channel is acceptable for the use case; in environments where it is not, the system can run with this pillar disabled (the core's security is unchanged).

4.2 Pillar 10: Self-Observation Resistance at the Failure Boundary

The greatest vulnerability in a cryptographic system often appears "at failure." Known attack families — fault attacks, side-channel attacks, downgrade attacks — all exploit the moment a system deviates from normal operation and its internal structure is exposed. This pillar is introduced as a cross-cutting design principle for systematically governing such exposure at the failure boundary.

The design principle reduces to a single sentence: **what is exposed when something breaks is only its "form"; the secret (the issuer key and the current value of dynamic internal state) does not leak even when it breaks**. Here "form" is the part that Kerckhoffs's principle presupposes to be public — concretely, the algebraic structures used, protocol identifiers, the referential structure of the data, and so on. These may be public without harming security, whereas the secret must always remain in the issuer's private half.

This pillar bundles the following four sub-functions in an integrated way.

Sub-function	Content	Corresponding existing technique
Desync detection	Detect seq #/nonce drift early	TLS 1.3 record sequence, QUIC packet number
Controlled graceful degradation	Stepwise retreat from the tail via back-pressure	TCP flow control, reactive streams
Internal-state leak resistance	Even on break, the core's current state does not emerge	Side-channel masking, fault-injection countermeasures
Observation-evasion representation	The same pattern cannot be captured twice via screenshot/OCR	Moving-target defense

Existing research on fault tolerance, side-channel countermeasures, and ratchets each has its own accumulated body of work, but a frame that bundles them under a single design principle has few precedents. The originality of this pillar is that it codifies Kerckhoffs's "separation of form and secret" as an implementation discipline **at the concrete, operable scene of the failure boundary**.

4.3 Pillar 3: Automatic Branching on Failure in the Tally Operators +/-

For the aggregation of digital signatures and the verification of relationships among multiple certificates, many existing techniques exist: BLS aggregate signatures (Boneh–Lynn–Shacham, 2004) [2], cryptographic accumulators, the presentation mechanism of W3C Verifiable Credentials, Myers diff, Merkle DAG diff, and so on. Each evolved for its own purpose, but from the standpoint of an operational authenticity-judgment UX they are not necessarily provided in an integrated form.

human cord integrates these into **a single operator algebra**. Specifically, the **+** operation expresses BLS aggregation, accumulator membership proof, and VC presentation as one and the same operation, while the **-** operation is defined as a difference-presentation mode that automatically falls back when **+** fails. The design in which **a failed operation returns not an error but another piece of useful information (a difference statement)** belongs to a rare class for a cryptographic protocol.

The operational significance of this pillar is clear. In ISMS audits, what was previously a binary judgment of "is this single signature verifiable?" is extended into a descriptive capacity of **"the relationships among multiple certificates can also be verified."** In practical UX, a verifier always obtains either "a single merged sheet upon match" or "a comparison display with the difference made explicit," so a state of "cannot judge" cannot arise in principle.

4.4 Pillar 7: Physical-Layer Cross-Modal Signal (the Visual Channel)

A cryptographic signal that survives media round-trips such as printing, scanning, and photography sits at the intersection of the EURion constellation, TEMPEST, Li-Fi, and the like, and is a relatively young area for mainstream cryptographic engineering. No practically deployed standard exists at present.

For this pillar, human cord began a **minimal implementation of the protocol layer** in Phase 2. There are two design points. First, the locus of guarantees is clearly separated: **a smartphone's AI image analysis serves as the "eye" (robustness against distortion, lighting, and partial occlusion), while the cryptographic guarantees (authenticity, tamper detection, freshness) are borne by the cord side**. Advances in AI do not take over the guarantees. Second, on the premise of the fact that **an optical channel by itself cannot prevent replay** (an image of a photographed screen can be re-displayed and re-photographed to duplicate it), replay prevention is placed in the "protocol," not the "channel." Concretely, it combines one-time-use verification of a per-issuance unique nonce (the chain-tip hash), a freshness window based on a public time axis, and the "smoke" (Pillar 8) that rises upon detection of tampering or re-presentation.

This vertical slice is expected to obtain its first application in the concrete operational field of certificate issuance in BiosGuide; physical adapters such as actual pixel rendering, error-correcting

codes, and camera/AI extraction remain as continuing work.

5. Connections to Existing Research

Each pillar of human cord has strong connections to existing cryptographic research, standards, and implementations. The novelty lies not in inventing individual techniques but in the way they are combined and in the cross-cutting design principles layered on top. This chapter presents the mapping to existing research carried out to avoid reinventing the wheel, and our positioning within a higher-level research area.

5.1 Ten Pillars × Existing Techniques (Compressed Matrix)

Pillar	Principal existing techniques	Standardization status
1 Zodiac multi-axis key	KDF / HKDF / ChaCha	NIST SP 800-108 / RFC 5869
2 Landscape blending	FPE / steganography	NIST SP 800-38G (FPE)
3 Tally operators	BLS aggregate sig / accumulator / VC	IETF draft-irtf-cfrg-bls-signature
4 Issuer's private half	PKI X.509 / threshold sig / Shamir SS	RFC 5280 / ISO/IEC 11770
5 Living operator	Signal Double Ratchet / sponge	IRTF CFRG / Signal spec
6 Subliminal channel	Simmons subliminal channel (1983)	(standardization gap)
7 Physical-layer signal	Physical-layer security / EURion / Li-Fi	IEEE 1900 series / 802.15.7
8 Active response	Cryptographic tripwire / HSM tamper / transparency log	FIPS 140-3 L4 / RFC 6962
9 Egg-flow architecture	AES-GCM / ChaCha20-Poly1305 / TLS 1.3	RFC 8446 / NIST SP 800-38D
10 Failure boundary	Side-channel countermeasures / MTD	(cross-cutting; no integrated standard)

5.2 Positioning Within a Higher-Level Research Area

human cord as a whole can be positioned as one implementation of the active research area of **Moving Target Defense (MTD) cryptography**. MTD has accumulated research centered on the DARPA Moving Target program (since 2010) and NIST SP 800-160 Vol. 2 (Systems Security Engineering), but as of 2026 no general-purpose standard has yet been established. This document positions human cord as a candidate standard proposal against that gap.

5.3 Making Existing vs. Novel Explicit

To make the scope of our novelty claims explicit, we organize the contribution category of each pillar as follows. Pillars 1, 4, and 9 adopt existing techniques as-is, and we claim no novelty (reinvention avoidance). Pillars 2, 5, 7, and 8 are positioned as derivations/extensions of existing techniques. Pillars 3, 6, and 10 carry a commensurate room for novel contribution, as detailed in § 4. And we present, as another contribution of this document, the integrating frame itself that ties the ten pillars into a single blueprint as five domains plus one cross-cutting layer.

6. Roadmap and Current Status

The human cord project is organized into six phases, from documenting the design through to standardization. Each phase has an independent exit deliverable and proceeds in an incrementally verifiable form. This chapter presents the overall roadmap and the several target specifications on the way to the final destination of international standardization.

6.1 Six-Phase Roadmap

Phase	Exit	Status
0 Document the foundation	Dream log + soul memo + architecture diagram + survey + this white paper	Complete
1 Minimal Node.js POC	A working minimal human cord	10 pillars + public-key layer (Ed25519) + adoption API, zero dependencies, 119 tests
2 Extension to the physical layer	A human cord that survives printing and photography	Started (Pillar 7 visual/audio dual carriers + RS error correction; physical adapters ongoing)
3 Concept white paper v0.2	A 5-page bilingual specification	This document = being finalized
4 BiosGuide integration	Embedded into certificate issuance + audit logs	Design / adoption surface / integration draft started (production rollout separate)
5 Approach to Blancco	Begin dialogue with technical staff	Not started
6 Standardization	IACR ePrint → SCIS → international conferences → IETF/NIST → ISO/IEC	Not started

6.2 Standardization Targets

Target specification	Relevant pillars	Estimated landing
ISO/IEC 18033 (encryption algorithms)	Pillars 5, 6, 9 integrated	5–10 years
ISO/IEC 29192 (lightweight cryptography)	BiosGuide IoT application	3–7 years
ISO/IEC 19772 (authenticated encryption)	Pillar 3 + Pillar 9	3–5 years
ISO/IEC 27002 annex	Whole-system operational guidance	3–5 years
IETF RFC (via IRTF CFRG)	Individual specs for Pillars 5, 6, 8	2–4 years

Standardization proceeds in stages along the layers of § 3.5: first propose layer (I), the symmetric issuing core (Pillars 1/5/9/10), to the CFRG as the minimal kernel; then layer (II), the public-verification layer (profiling the existing Ed25519 / RFC 8032 standard); then stack layer (III), the distinctive value (Pillar 3 / Pillar 4 threshold / Pillar 7 carrier), as extension specifications. Layer (IV) is deferred as future extensions and operational guidance (the ISO/IEC 27002 annex).

Shortest route: IRTF CFRG (layer I core) → IETF RFC → ISO adoption (extending through layer III in stages)

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Note: each entry is based on a publicly known standard or paper, but edition numbers and years of publication will be finally cross-checked before distribution. Augmentation to roughly 40

entries is planned upon completion of Phase 3.

Appendix A: Terminology Correspondence (Metaphor ↔ Cryptographic Engineering)

Metaphor (human cord)	Cryptographic engineering
Egg	AEAD record / sealed box
Egg chain	Hash-linked record stream
Egg core	Stateful operator (ratchet)
Eye-glyph	Selectively format-preserved substituted character
Eye-glyph noise	Subliminal channel payload
Landscape blending	Non-substituted plaintext context
Zodiac multi-axis	Multi-dimensional KDF input
Pass token	Issuer-specific verification key
Tally (split tally)	Issuer's private half (cf. PKI private key)
Smoke	Tamper-evident broadcast signal
Private half	Private half of an asymmetric pair
Failure boundary	Failure-mode operational boundary
Rhythm break	seq #/nonce desync
A tail egg drops	Back-pressure-induced graceful degradation

Revision History

- 2026-05-28 Created as Phase 0 Task #6 (Japanese edition); § 1–6 finalized in white-paper prose.
- 2026-05-29 Established as an external deliverable under `docs/`. Finalized cover elements (author name and contact are placeholders to be inserted just before Phase 3 publication; the rest are fixed), updated status to current state, promoted § 4.4 Pillar 7 to a Phase 2 protocol-layer POC, converted the architecture diagram to Mermaid, and expanded references from 12 to 35. English parallel edition produced.
- 2026-05-30 Added § 3.4 "Two Layers of Verification (issuer-mediated + public)" (Pillar 4 public-key layer, Ed25519 = offline third-party verification with only the public key; Web Crypto interop confirmed = serverless verification page feasible). Updated § 6.1 roadmap to the current state (public-key layer, adoption API, 119 tests, Phase 4 design/draft started).

- 2026-05-30 Added § 3.5 "Layering Toward Standardization: Core and Extensions." A threat→mechanism→property teardown of the ten pillars sorts them into four layers: (I) symmetric issuing core / (II) public-verification layer / (III) distinctive value / (IV) future extensions, cross-cutting principle, and operational conventions (the substantive specification core is roughly five functions). Reframed the § 6.2 standardization targets as a staged process along these layers (the ten-pillar conceptual map = Appendix A is preserved).
- 2026-05-31 Added to § 4.1 an "operational positioning (an important reservation)" for Pillar 6 (subliminal channel). Because covert channels are treated with suspicion in audit/standardization contexts, the pillar is quarantined out of the verifiable core (§ 3.5 layers I–III) as a future extension (layer IV) and can be run disabled (core security unchanged).